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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

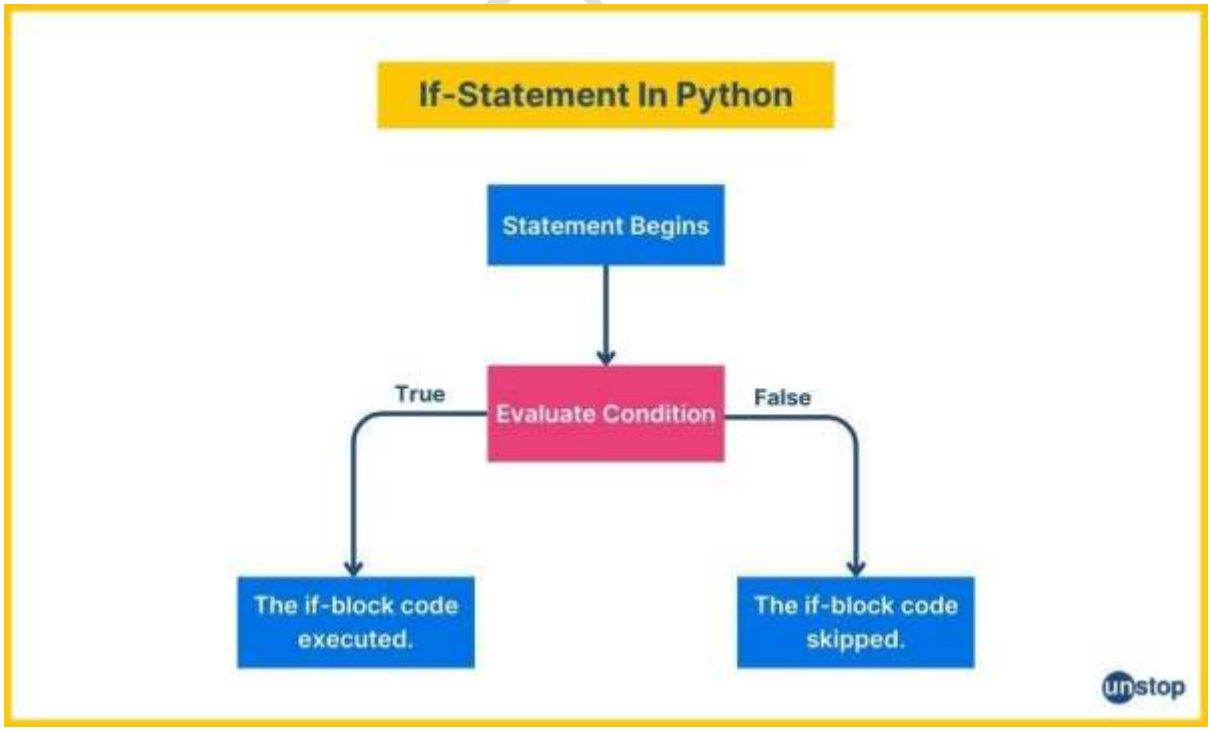
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 20: Conditional Statements in Python



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what conditional statements are.
- ✅ Learn how computers make decisions.
- ✅ Use if, if-else, and if-elif-else statements.
- ✅ Write Python programs using conditions.
- ✅ Solve simple real-life problems with decision-making.

What are Conditional Statements?

A **Conditional Statement** helps the computer **make decisions**.

It checks whether a condition is **True** or **False** and then performs the appropriate action.

Real-Life Example

Imagine a traffic signal:

- 🟢 **Green** → Go
- 🟡 **Yellow** → Wait
- 🔴 **Red** → Stop

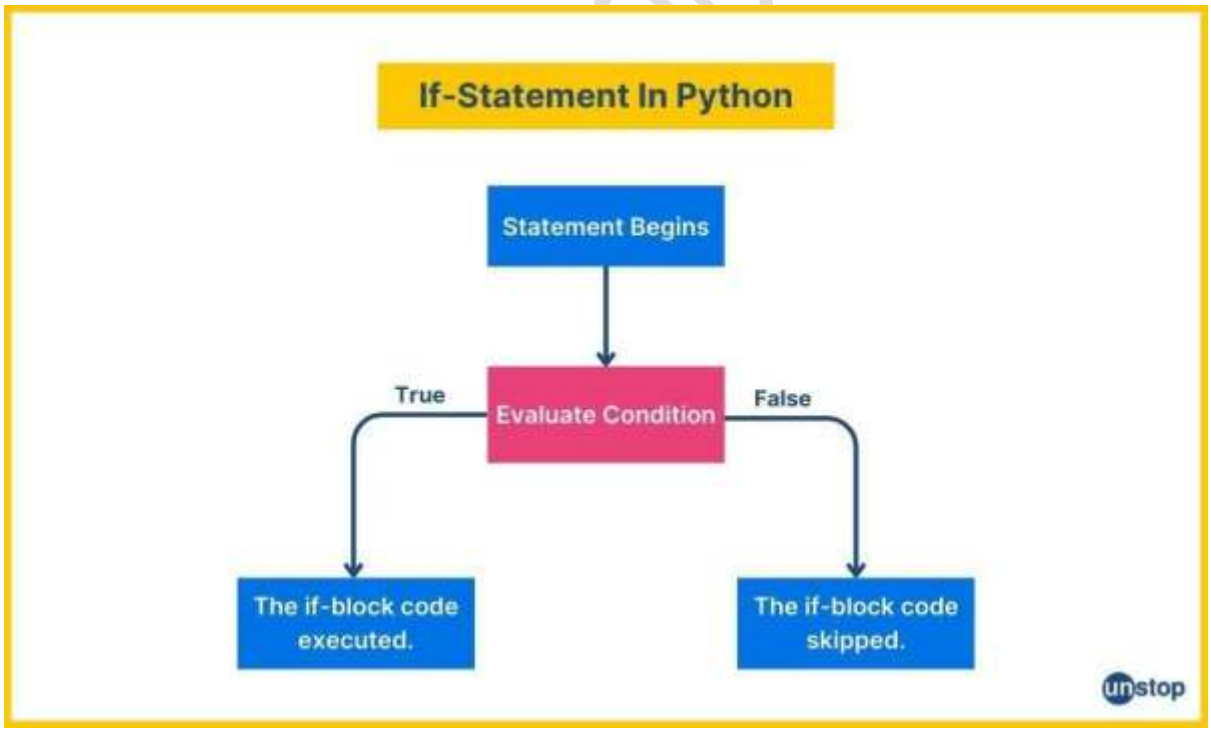
The driver makes a decision based on the signal.

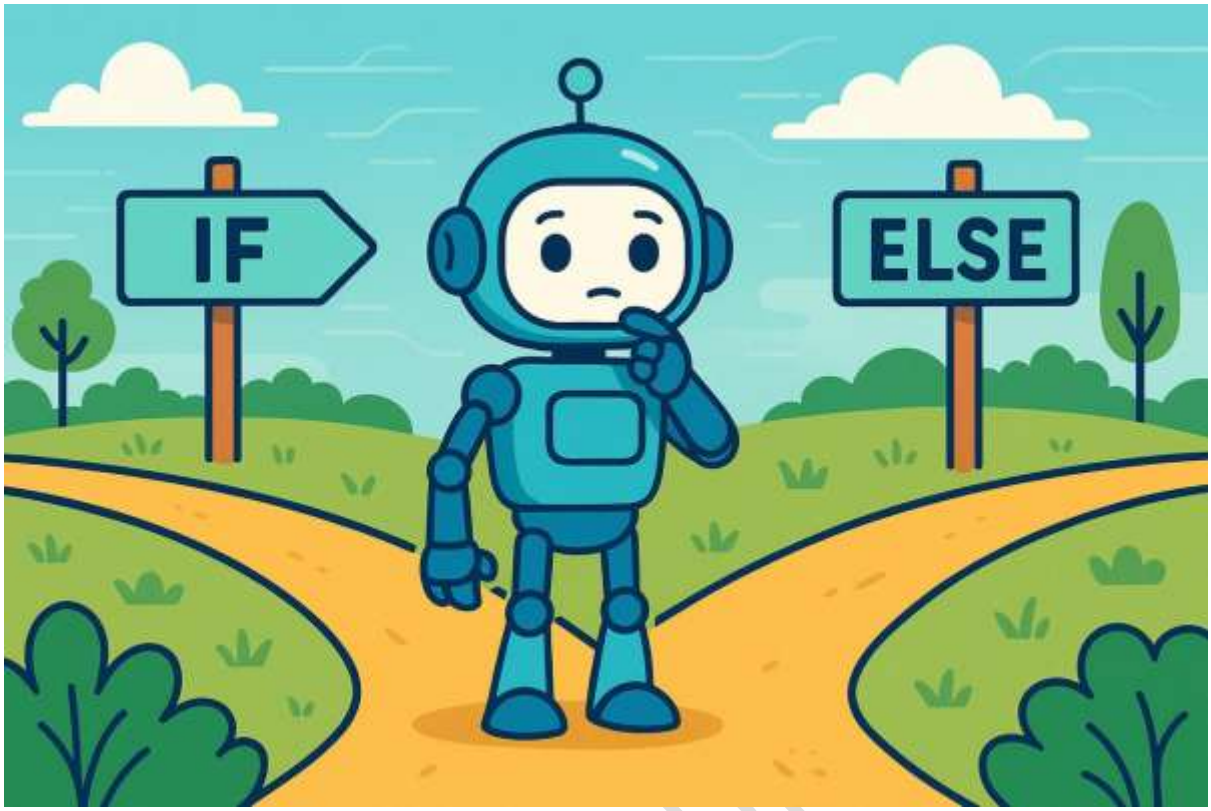
Similarly, Python makes decisions using **conditional statements**.

Decision Making in Python

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STOP
READY
GO


If-Statement In Python
Statement Begins
Evaluate Condition
True
False
The if-block code executed.
The if-block code skipped.
unstop



Why Do We Use Conditional Statements?

Conditional statements help us:

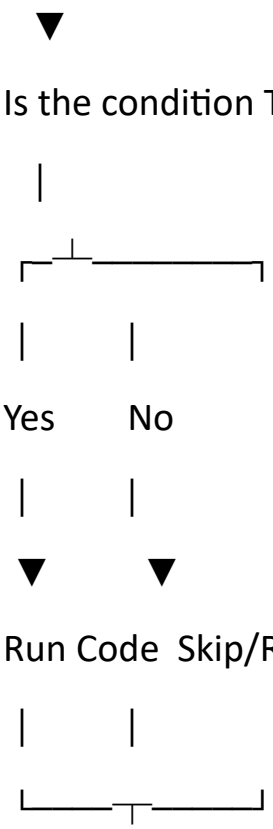
- Decide between two or more choices.
- Check marks and grades.
- Verify passwords.
- Compare numbers.
- Build games and interactive programs.
- Create smart applications.

Decision Flow

Start

|

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Types of Conditional Statements

Statement	Purpose
if	Executes code if the condition is True
if...else	Chooses between two actions
if...elif...else	Chooses from multiple actions
Nested if	Uses an if statement inside another if

The if Statement

The if statement executes code **only when the condition is True**.

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Syntax

```
if condition:
    statement
```

Example 1

```
age = 15

if age >= 10:
    print("You are eligible.")
```

Output

```
You are eligible.
```

Example 2

```
marks = 80

if marks >= 35:
    print("Pass")
```

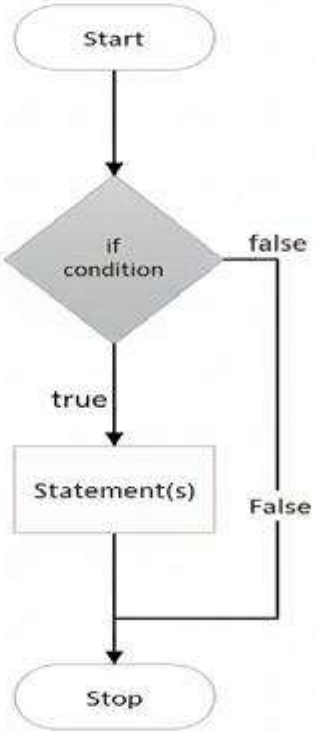
Output

```
Pass
```

IF Statement

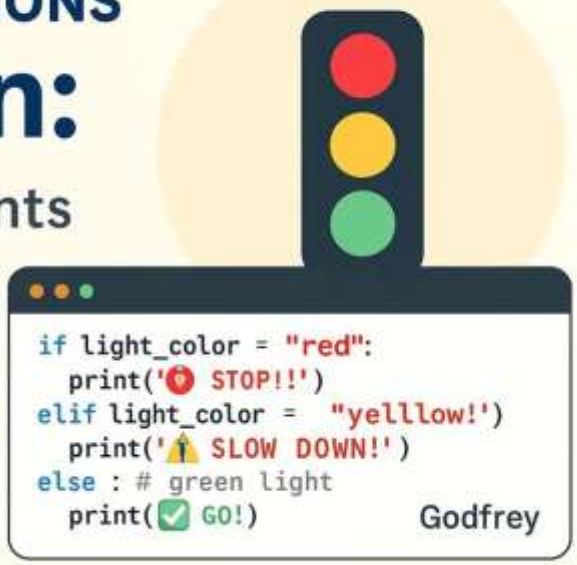
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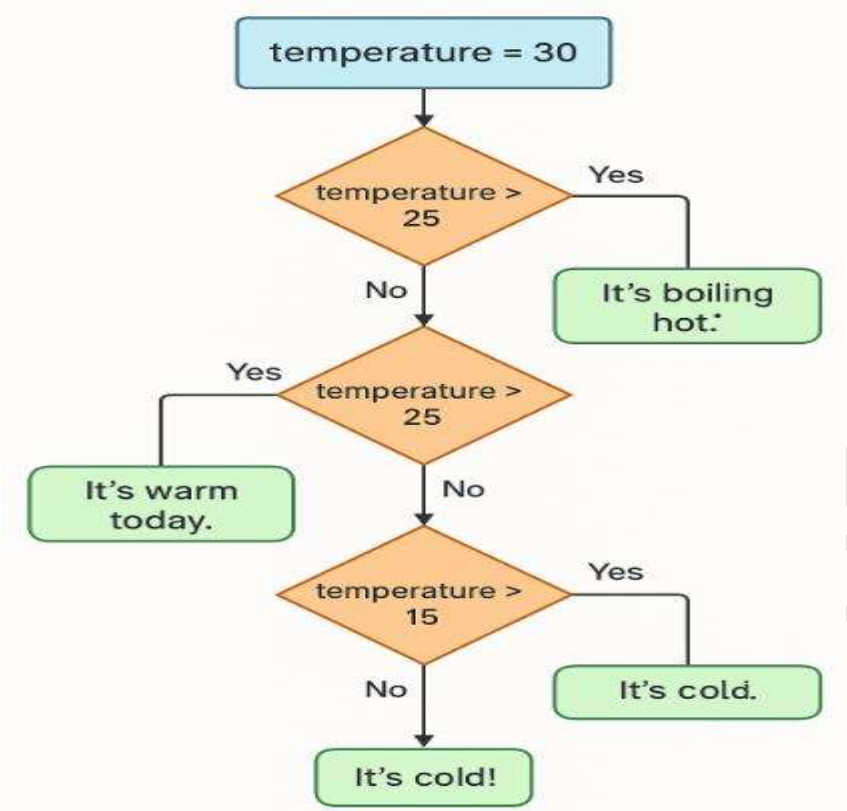
Flowchart of if statement



MAKING DECISIONS IN Python:

If/Else Statements Made Simple





The if...else Statement

The else block runs when the condition is **False**.

Syntax

```
if condition:
    statement
else:
    statement
```

Example

marks = 25

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```
if marks >= 35:
   print("Pass")
else:
   print("Fail")
```

Output

Fail

Example 2

```
number = 8

if number % 2 == 0:
   print("Even Number")
else:
   print("Odd Number")
```

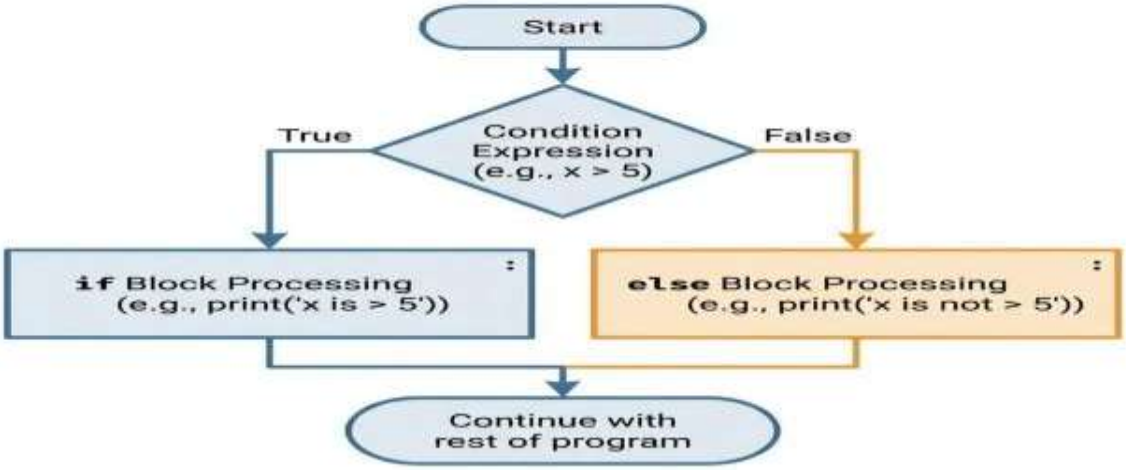
Output

Even Number

IF-ELSE Flowchart

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Basic `if-else` Statement Flow



Condition is True

```
number = 10
if number > 0:
    # code

else:
    # code

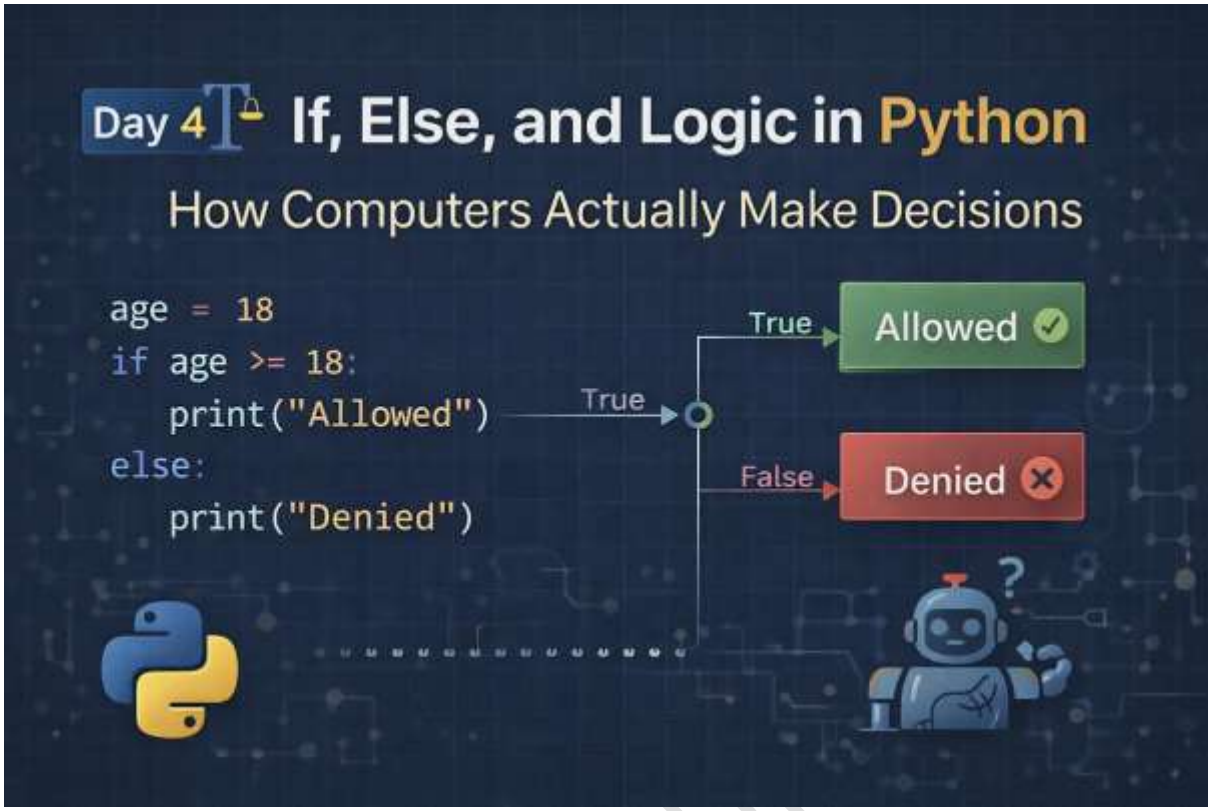
# code after if
```

Condition is False

```
number = -5
if number > 0:
    # code

else:
    # code

# code after if
```



The if...elif...else Statement

Use elif to check **multiple conditions**.

Syntax

```
if condition1:
    statement
elif condition2:
    statement
else:
    statement
```

Example

```
marks = 92
```

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```
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

Output

Grade A

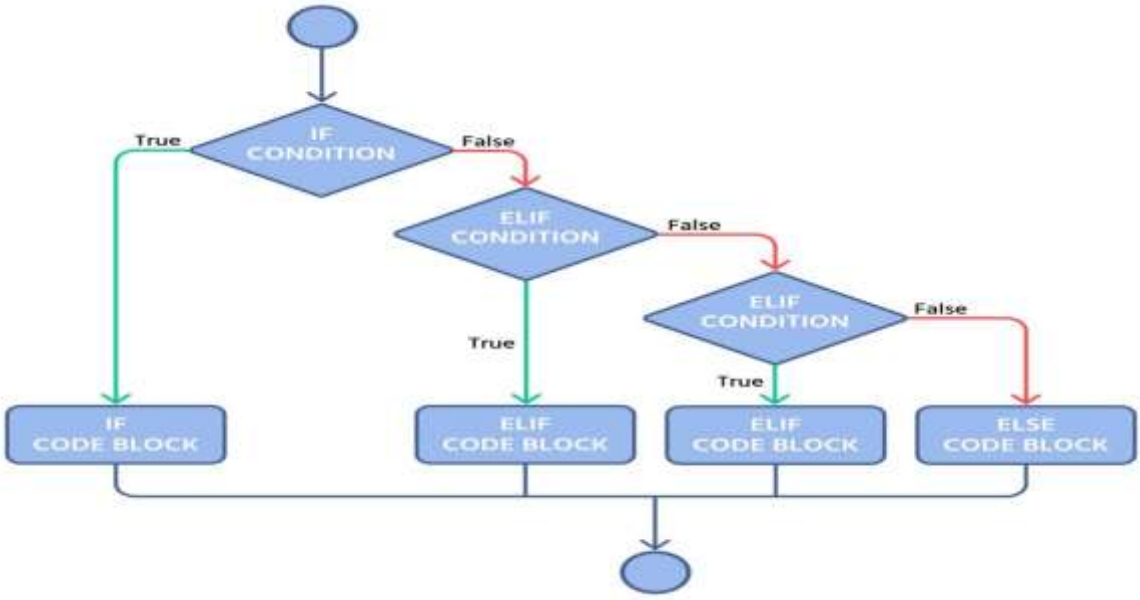
Grade Chart

Marks	Grade
90–100	A
75–89	B
50–74	C
Below 50	Fail

Grade Decision

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Nested if Statement

A **Nested if** means an if statement inside another if.

Example

```
age = 18

has_id = True


if age >= 18:
    if has_id:
        print("You can enter.")
```

Output

```
You can enter.
```

Using Comparison Operators

Operator	Meaning
==	Equal
!=	Not Equal
>	Greater Than
<	Less Than
>=	Greater Than or Equal
<=	Less Than or Equal

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Practical Session

Practical Activity 1

Check Pass or Fail

```
marks = int(input("Enter Marks: "))

if marks >= 35:
    print("Pass")
else:
    print("Fail")
```

Practical Activity 2

Check Even or Odd

```
number = int(input("Enter Number: "))

if number % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Practical Activity 3

Find Greater Number

```
a = int(input("First Number: "))
b = int(input("Second Number: "))

if a > b:
    print(a, "is Greater")
else:
```

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```
    print(b, "is Greater")
```

Practical Activity 4

Voting Eligibility

```
age = int(input("Enter Age: "))

if age >= 18:
    print("Eligible to Vote")
else:
    print("Not Eligible")
```

Practical Activity 5

Grade Calculator

```
marks = int(input("Enter Marks: "))

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

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Practical Activity 6

Positive or Negative Number

```
number = int(input("Enter Number: "))


if number >= 0:
    print("Positive Number")
else:
    print("Negative Number")
```

Practical Activity 7

Check Password

```
password = input("Enter Password: ")


if password == "python123":
    print("Access Granted")
else:
    print("Access Denied")
```

Practical Activity 8

Largest of Three Numbers

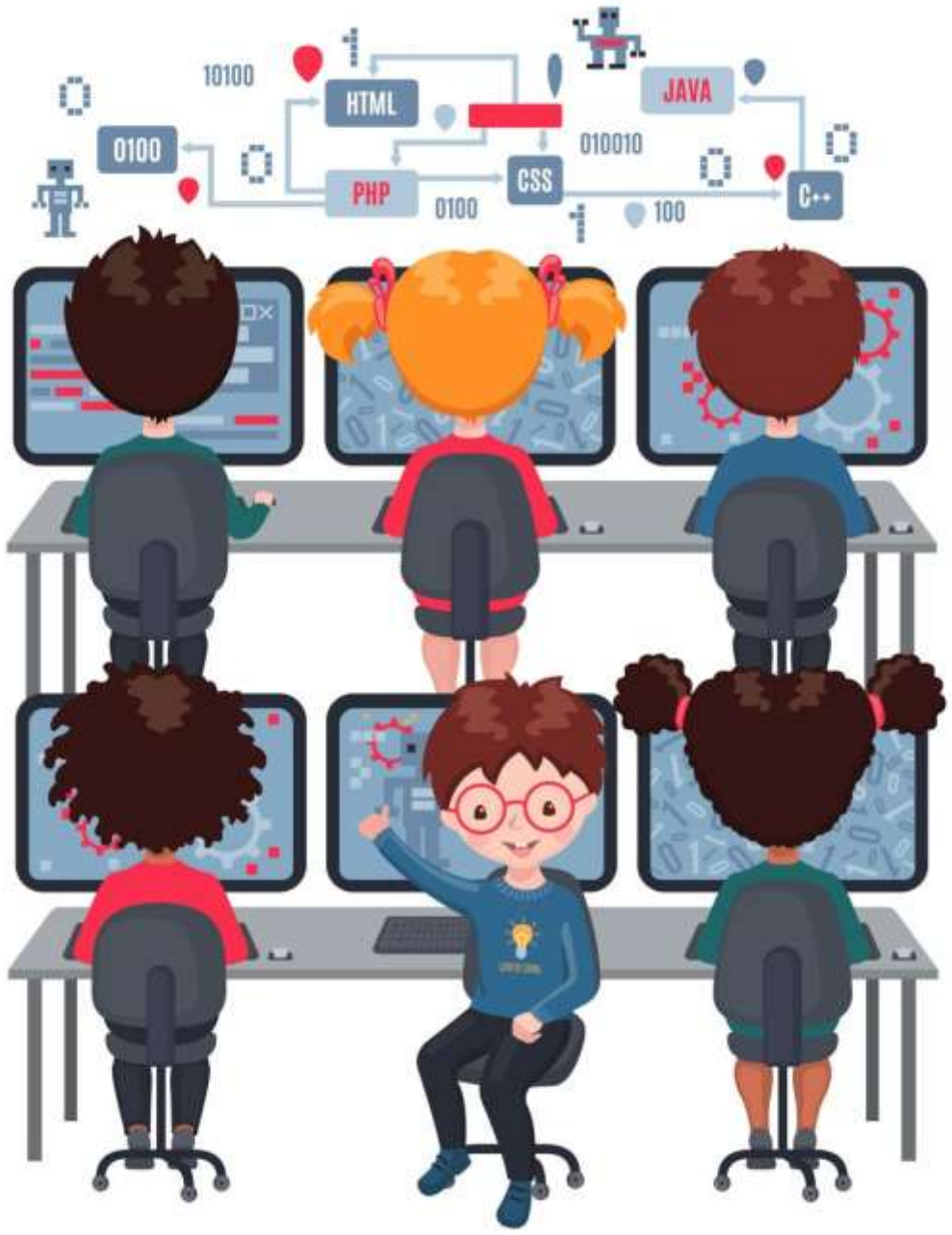
```
a = int(input("First Number: "))
b = int(input("Second Number: "))
c = int(input("Third Number: "))
```

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```
if a >= b and a >= c:
    print(a, "is Largest")
elif b >= a and b >= c:
    print(b, "is Largest")
else:
    print(c, "is Largest")
```

Classroom Coding Activity

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10100
HTML
JAVA
0100
010010
PHP
0100
CSS
100
C++





Activity 1

Write a program to check whether a student passed or failed.

Activity 2

Write a program to check whether a number is positive or negative.

Activity 3

Write a program to compare two numbers.

Activity 4

Write a program to display grades based on marks.

Real-Life Applications

Situation	Conditional Statement
ATM PIN Verification	if...else
School Result	if...elif...else
Online Login	if
Traffic Signal	if...elif...else
Weather App	if...else
Game Winner	if

Common Mistakes

❌ **Missing Colon (:)**

Wrong

```
if marks >= 35
   print("Pass")
```

✅ **Correct**

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```
if marks >= 35:
   print("Pass")
```

❌ **Incorrect Indentation**

Wrong

```
if marks >= 35:
print("Pass")
```

✅ **Correct**

```
if marks >= 35:
   print("Pass")
```

❌ **Using = Instead of ==**

Wrong

```
if marks = 35:
```

✅ **Correct**

```
if marks == 35:
```

Best Practices

- ✓ Use meaningful variable names like marks, age, and password.
- ✓ Always indent the code inside if, else, and elif.
- ✓ Write clear and simple conditions.
- ✓ Test your program with different inputs.

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Keywords

- **Condition** – A True or False test.
- **Conditional Statement** – A decision-making statement.
- **if** – Executes code when a condition is True.
- **else** – Executes code when the condition is False.
- **elif** – Checks another condition if the previous one is False.
- **Nested if** – An if statement inside another if.

Lesson Summary

- Conditional statements help computers make decisions.
- The if statement runs code only when a condition is True.
- The if...else statement chooses between two actions.
- The if...elif...else statement handles multiple choices.
- Nested if statements allow more detailed decision-making.
- Conditional statements are used in many real-life applications such as login systems, grading, and games.

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Quiz

Multiple Choice Questions

1. Which keyword is used to make a decision in Python?

A) for

B) if

C) while

D) print

✅ **Answer: B**

2. Which statement is used when there are multiple conditions?

A) while

B) elif

C) input

D) print

✅ **Answer: B**

3. What will this code print?

```
marks = 40

if marks >= 35:
    print("Pass")
else:
    print("Fail")
```

A) Fail

B) Pass

C) Error

D) Nothing

✅ **Answer: B**

4. Which comparison operator checks equality?

A) =

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B) ==

C) !=

D) >=

✅ **Answer: B**

5. Why is indentation important in Python?

A) It changes text color.

B) It tells Python which code belongs to a block.

C) It makes the program faster.

D) It is optional.

✅ **Answer: B**

Homework

1. Write a program to check whether a number is even or odd.
2. Write a program to find the greater of two numbers.
3. Write a program to display grades based on marks.
4. Write a program to check voting eligibility using age.
5. Write a program to check whether a number is positive or negative.
6. Write a program to find the largest of three numbers.

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